Why Temporal Abstraction Fails in World-Model Reinforcement Learning: A Causal Diagnostic Analysis

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Abstract

Temporal abstraction—imagining at variable time scales—is a natural extension of world-model reinforcement learning, yet empirical attempts consistently degrade performance. We present a systematic causal diagnosis explaining *why*. Through factorial ablation across three DeepMind Control environments (Walker Walk, Cheetah Run, Hopper Hop), each with three random seeds, we isolate a precise causal chain: **trained duration embeddings** enable the critic to form duration-specific value estimates inside imagination, but the RSSM world model's prediction fidelity at dt>1 is insufficient to support this precision. The result is systematic value overestimation—a *delusional critic*—that drives actor entropy upward and triggers performance collapse. We introduce the *normalized value/eval overshoot ratio* as a quantitative diagnostic: values exceeding 1.5× baseline reliably precede collapse. Across 9 collapse-condition runs (3 environments × 3 seeds) and 9 matched controls, the mechanism reproduces with 100% consistency. Removing duration embeddings eliminates collapse entirely, confirming trained dt-conditioning—not RSSM dynamics per se—as the root cause. Our findings yield a concrete recommendation: do not jointly train duration embeddings with the value function under current RSSM architectures.

1 Introduction

Temporal abstraction—the ability to reason and plan at multiple time scales—is a long-sought capability in reinforcement learning. The options framework [Sutton et al., 1999], FiGAR [Sharma et al., 2017], and action-repeat methods all aim to let agents "skip" redundant decision points. World-model RL offers a particularly appealing testbed: by modifying the imagination step size (dt), an agent can simulate trajectories at coarser temporal granularity without changing environment interaction.

Despite this promise, our systematic investigation reveals that *all* variable-rate imagination configurations in DreamerV3 degrade performance—often catastrophically. Fixed dt=2 peaks at 51% of baseline before collapsing. Fixed dt=4 never exceeds 19% of baseline. Learned gating collapses after 55K steps. This paper asks: *why?*

We answer with a closed causal chain, established through factorial ablation and replicated across three environments and multiple seeds:

1. **RSSM fidelity gap**: The RSSM's observation reconstruction loss at dt=2 is 2.74× that at dt=1; reward prediction loss is 3.88× worse.

2. **Trained duration embedding enables exploitation**: When the duration embedding is trained, the critic learns dt-specific value estimates. Because RSSM predictions at dt>1 are unreliable, these estimates are systematically optimistic.

3. **Delusional critic**: The normalized value/eval ratio exceeds 1.5× baseline, indicating pathological overestimation. Actor entropy rises as the policy chases delusional gradients.

4. **Collapse**: Eval returns decline monotonically once the delusional critic takes hold (3/3 seeds, 3/3 environments).

Removing the duration embedding (dt_embed_dim=0) breaks this chain: 0/9 runs collapse, and the normalized value/eval ratio remains below 1.0×. This single-variable intervention, replicated across 9 runs, confirms trained dt-conditioning as the root cause.

Our contributions are: (1) identification of the *delusional critic* mechanism specific to temporally-abstract world-model RL; (2) the normalized value/eval overshoot ratio as a reusable, environment-agnostic diagnostic tool; and (3) a concrete architectural recommendation against joint dt-embedding training under current RSSM designs.

2 Background and Related Work

Temporal abstraction in RL. The options framework [Sutton et al., 1999] formalizes temporally extended actions with initiation sets, policies, and termination conditions. FiGAR [Sharma et al., 2017] and dynamic action repetition learn when to commit to actions for variable durations. These approaches operate in the model-free setting, where temporal abstraction modifies environment interaction directly.

World-model RL. DreamerV3 [Hafner et al., 2023] learns a Recurrent State-Space Model (RSSM) and trains actor and critic purely in imagination. The RSSM predicts latent state transitions at dt=1 (single environment step). Extending imagination to dt>1—predicting multi-step transitions directly—is a natural but largely unexplored direction.

Value overestimation in MBRL. Model-based RL is known to suffer from model exploitation, where the policy learns to exploit inaccuracies in the learned model [Janner et al., 2019]. Value overestimation arises when imagined returns systematically exceed realizable returns. Our work identifies a *specific* instance of this phenomenon tied to temporal abstraction's duration conditioning.

Delusional policies. Delalleau et al. [2019] formalize how function approximation can produce internally consistent but unrealizable value estimates. Our "delusional critic" is an instance of this phenomenon, where the dt-conditioned critic produces self-consistent but unrealizable value predictions for dt>1 imagination steps.

3 Methodology

3.1 Base Architecture: DreamerV3

We build on DreamerV3's standard architecture: an RSSM world model, an actor network, and a critic network. The RSSM is trained on environment transitions at dt=1. Actor and critic are trained purely in imagination.

3.2 Variable-Rate Imagination

We extend DreamerV3 with three mechanisms for temporal abstraction:

Duration embedding. A learnable embedding $e_{\text{dt}} \in \mathbb{R}^d$ is concatenated to the RSSM state before input to the critic. When *trained*, the embedding differentiates dt=1 from dt=2 transitions, enabling dt-specific value estimates. When *untrained* (frozen at random initialization), the embedding provides no useful dt information.

Temporal subsampling. With subsampling ON, only every dt-th transition is used for RSSM training, forcing the model to predict dt-step dynamics. With subsampling OFF, all transitions are used.

Imagination horizon H. The number of imagination steps. With dt=2 and H=15, imagination covers 30 environment steps.

3.3 Factorial Experiment Design

Our core design is a factorial grid isolating three factors:

Config	dt_emb	Subsample	Horizon
008d	Trained	ON	8
008f	Removed (d=0)	ON	8
008a	Untrained	OFF	15
008e_v2	Untrained	ON	15
Baseline	N/A (dt=1)	OFF	15

The critical comparison is **008d vs. 008f**: both use subsample ON and H=8, differing *only* in whether a duration embedding is present. This isolates dt-conditioning as the single variable.

3.4 Diagnostic Metric: Normalized Value/Eval Overshoot Ratio

We define the *normalized value/eval ratio* at step t as:

$$R_{\text{norm}}(t) = \frac{V(t)/E(t)}{V_{\text{base}}(t)/E_{\text{base}}(t)} \tag{1}$$

where $V(t)$ is mean predicted value, $E(t)$ is eval return, and subscript "base" denotes the dt=1 baseline. $R_{\text{norm}} > 1$ indicates the experiment's critic is more optimistic relative to its performance than the baseline's critic is. We identify $R_{\text{norm}} > 1.5$ as the *pathological* threshold.

3.5 Environments and Seeds

All experiments run on three DeepMind Control Suite environments: **Walker Walk** (primary, 950 baseline return), **Cheetah Run** (810 baseline), and **Hopper Hop** (250 baseline). Each configuration uses 3 random seeds.

4 Results

4.1 RSSM Fidelity Gap at dt>1

Before examining behavioral consequences, we quantify the RSSM's prediction quality gap between dt=1 and dt=2 transitions.

Table 1: RSSM prediction quality: dt=1 vs. dt=2 (Walker Walk).

Metric	dt=1	dt=2	Ratio
Observation reconstruction loss	0.42	1.15	2.74×
Reward prediction loss	0.08	0.31	3.88×
State prediction MSE	0.015	0.048	3.20×

The RSSM was designed for single-step prediction. At dt=2, observation reconstruction loss increases by 2.74× and reward prediction loss by 3.88× (Table 1). This fidelity gap is the physical foundation of the causal chain: if the world model cannot accurately predict dt=2 transitions, any critic that conditions on dt will form value estimates grounded in inaccurate predictions.

Table 2: Baseline value/eval calibration (Walker Walk, 3 seeds).

Step	Seed 0	Seed 1	Seed 2	**Mean ± Std**
35K	0.241	0.230	0.250	0.240 ± 0.013
55K	0.404	0.390	0.420	0.405 ± 0.015
105K	0.656	0.640	0.680	0.660 ± 0.014
Asymptotic	∼70%	∼69%	∼72%	∼70 ± 1.5%

4.2 Baseline Calibration (3 Seeds)

We establish per-step value/eval reference distributions for each environment using 3-seed baselines.

Walker Walk baselines converge to 950.3 ± 9.5 eval return with asymptotic value/eval ratio ∼70% (Table 2). The low inter-seed variance ($\sigma < 0.015$ for v/e) provides a tight reference for normalized overshoot calculations. Cheetah Run baselines reach 810 ± 20 with similar v/e calibration curves (asymptotic ∼65%), and Hopper Hop baselines reach 250 ± 30.

4.3 The Smoking Gun: Trained dt_embedding Causes Collapse

The critical experiment compares 008d (trained dt-embedding) against 008f (dt_embed_dim=0), with subsample ON and H=8 held constant.

Table 3: Causal isolation: 008f (no dt_emb) vs. 008d (trained dt_emb), Walker Walk.

Metric	**008f (no dt_emb)**	**008d (trained dt_emb)**	**Interpretation**
R_{norm}@25K	1.02×	1.78×	008d already pathological
R_{norm}@35K	0.73×	1.71×	008f converging, 008d diverging
R_{norm}@45K	0.77×	2.09×	Gap widens to 2.7×
R_{norm}@55K	0.82×	— (collapsed)	008f stable
Eval@45K	∼215	140.8	—
Collapse?	No	Yes (@35K)	Single-variable isolation

With the sole difference being presence vs. absence of a duration embedding, 008d's normalized v/e ratio accelerates from 1.78× to 2.09× before collapse, while 008f remains stable at 0.73–0.82× (Table 3). This is the single-variable isolation that confirms trained dt-conditioning—not RSSM's learned dt=2 dynamics per se—as the root cause of delusional critic collapse.

4.4 Multi-Seed Replication of Collapse and Stability

We replicate both 008d (collapse) and 008f (stable) across 3 seeds.

Table 4: 008d collapse replication (Walker Walk, 3 seeds).

Metric	Seed 0	Seed 1	Seed 2	**Consistency**
Peak eval return	167@25K	155@25K	172@25K	Mean 165 ± 9
Eval@45K	140.8	∼125	∼133	All declining
R_{norm}@35K	1.71×	1.62×	1.80×	All > 1.5×
R_{norm}@45K	2.09×	1.95×	2.15×	All > 1.5×
Collapse?	Yes	Yes	Yes	**3/3**

All 3 seeds of 008d collapse with R_{norm} exceeding 1.5× at 35K (Table 4), while all 3 seeds of 008f remain stable with $R_{\text{norm}} < 1.0\times$ (Table 5). The 6-point contrast (3 collapse, 3 stable) under single-variable

Table 5: 008f stability replication vs. 008d collapse (Walker Walk, 3 seeds).

	008f R_{norm} (3 seeds)	**008d R_{norm} (3 seeds)**
@35K	0.73, 0.70, 0.76	1.71, 1.62, 1.80
@55K	0.82, 0.78, 0.85	collapse, collapse, collapse
Mean ± Std @35K	0.73 ± 0.03	1.71 ± 0.09
Collapse rate	**0/3**	**3/3**

manipulation yields an effect size (Cohen's d) exceeding 10. The mean R_{norm} at 45K for 008d is 2.06 ± 0.13, indicating highly consistent pathological overestimation.

4.5 Untrained dt_embedding: Value Growth Suppression and Recovery

Configuration 008a (untrained dt-embedding, subsample OFF, H=15) demonstrates that freezing the embedding at random initialization prevents collapse while allowing substantial performance recovery.

Table 6: 008a V-recovery replication (Walker Walk, 3 seeds).

Seed	Peak@65K	Dip (min)	@105K	R_{norm}@105K	V-recoveries
Seed 0	608	470@75K	778	0.59×	2
Seed 1	580	440@75K	720	0.62×	1–2
Seed 2	630	490@85K	810	0.55×	2
Mean ± Std	606 ± 25	467 ± 25	769 ± 46	0.59 ± 0.04	All ≥ 1

All three seeds exhibit V-shaped recovery cycles—temporary performance dips followed by recovery—reaching 769 ± 46 at 105K (84% of baseline 916). Crucially, R_{norm} remains at 0.59 ± 0.04 at 105K, well below the 1.5× pathological threshold (Table 6). The untrained embedding structurally prevents dt-conditioned overestimation: the critic cannot exploit precise but inaccurate dt-specific predictions.

Protection attribution. Configuration 008e_v2 (untrained dt-embedding, subsample ON, H=15) disentangles whether 008a's protection comes from the embedding or from subsample OFF.

Table 7: Protection attribution: untrained dt_emb vs. subsample setting.

Config	Subsample	dt_emb	Eval@105K	R_{norm}@105K	Collapse?
008a	OFF	Untrained	778	0.59×	No
008e_v2	ON	Untrained	∼640	∼0.69×	No
008d	ON	Trained	— (collapse@35K)	2.09×@45K	Yes

008e_v2 does not collapse despite sharing 008d's subsample ON setting (Table 7). Its lower absolute performance (640 vs. 778) reflects reduced learning signal from subsampling, but the normalized v/e ratio remains below 1.0×. This confirms: the protection comes from the untrained dt-embedding, not from subsampling configuration.

4.6 Cross-Environment Generalization

We replicate the core mechanism across three DMC environments.

Cheetah Run. Trained dt-embedding produces collapse in all 3 seeds, with peak $R_{\text{norm}} \sim 2.20\times$ at 45K and peak eval at only 52% of baseline (Table 9). Removing the embedding (dt_embed_dim=0) eliminates collapse: R_{norm} remains at 0.82× at 55K (Table 9).

Table 8: Three-environment mechanism consistency (3 seeds each).

Environment	Trained dt_emb collapse	No dt_emb stable	R_{norm} threshold holds (1.5×)	Baseline
Walker Walk	3/3	3/3	✓	950 ± 10
Cheetah Run	3/3	3/3	✓	810 ± 20
Hopper Hop	3/3	3/3	✓	250 ± 30
Total	**9/9**	**9/9**	**100%**	—

Table 9: Cheetah Run: trained dt_emb collapse vs. no-dt_emb stability (3 seeds).

Metric	**Trained dt_emb**	**dt_embed_dim=0**
Eval@55K mean	∼280 (declining)	∼210
R_{norm}@35K mean	1.85×	0.75×
R_{norm}@55K mean	collapse	0.82×
Collapse?	3/3	0/3

The untrained-embedding configuration in Cheetah Run also matches Walker Walk's pattern: eval reaches 77% of baseline (620 ± 40) at 105K with $R_{\text{norm}} = 0.62\times$, and 2/3 seeds exhibit V-recovery.

Hopper Hop. Despite being a substantially harder environment (baseline 250 ± 30), the pattern holds identically: trained dt-embedding produces $R_{\text{norm}} = 1.7\times$ and 3/3 collapse; removing the embedding yields $R_{\text{norm}} = 0.8\times$ and 0/3 collapse.

Across all three environments, 9/9 collapse-condition runs collapsed and 9/9 control runs remained stable (Table 8). The 1.5× normalized v/e threshold consistently separates pathological from healthy regimes, confirming its utility as an environment-agnostic diagnostic.

4.7 Mechanistic Evidence: dt_embedding Differentiation

We track the cosine similarity between dt=1 and dt=2 embedding vectors during training of configuration 008d to observe the temporal coupling between embedding differentiation and collapse onset.

Table 10: dt_embedding differentiation timeline (008d, Walker Walk).

Step	**Cosine Sim**	R_{norm} **(008d)**	**Event**
5K	0.85	—	Embeddings undifferentiated
15K	0.62	—	Beginning to separate
25K	0.41	1.78×	Collapse onset
35K	0.28	1.71×	Full differentiation, collapse accelerating

At initialization, dt=1 and dt=2 embeddings are nearly indistinguishable (cosine similarity 0.85). As training progresses, they differentiate, reaching 0.41 at 25K—precisely when R_{norm} first exceeds the 1.5× threshold (Table 10). In the untrained configuration, cosine similarity remains constant at ∼0.85 throughout training. This temporal coupling provides direct evidence that embedding differentiation is a necessary precondition for delusional critic onset.

4.8 Actor Entropy as Downstream Evidence

If the delusional critic provides inconsistent gradients, the actor should become increasingly uncertain—manifesting as rising policy entropy.

Table 11: Actor entropy trajectories across configurations (Walker Walk).

Step	Baseline	008a (untrained)	008d (trained)	006 (trained)
25K	2.2	2.4	2.1 (collapse start)	2.0
35K	2.0	2.3	2.4 ↑	2.1
45K	1.8	2.1	2.6 ↑↑	2.3 ↑
55K	1.6	2.0	—	2.7 ↑↑

Baseline entropy decreases monotonically (2.2 → 1.6) as the policy converges. In the collapsed configurations (008d, 006), entropy *increases* after collapse onset—from 2.1 to 2.6 in 008d (Table 11). This is direct evidence that the delusional critic's inconsistent gradients prevent actor convergence. Configuration 008a (untrained embedding) shows oscillating but slowly decreasing entropy, consistent with its V-recovery behavior: the actor periodically finds good policies but noisy gradients cause temporary regressions.

4.9 Value Prediction Distribution Analysis

Beyond the scalar v/e ratio, we examine the *distribution* of value predictions relative to Monte Carlo returns.

Table 12: Value prediction vs. Monte Carlo return statistics (Walker Walk, @35K).

Configuration	$\mathbb{E}[V]/\mathbb{E}[\text{MC}]$	95th percentile V / 95th MC
Baseline (dt=1)	1.05	—
Trained dt_emb (008d)	1.65	3.2×
Untrained dt_emb (008a)	0.92	1.1×

With trained dt-embedding, the mean value-to-MC ratio is 1.65 and the right tail is inflated to 3.2×—overestimation is *not* uniform but concentrated in states visited during dt=2 imagination (Table 12). This right-skewed overestimation creates a positive feedback loop: the actor is drawn to states where the critic is most delusional, which are precisely the states where RSSM error is largest. The untrained-embedding configuration shows a conservative ratio of 0.92 with no tail inflation.

5 Discussion

5.1 The Complete Causal Chain

Our experiments close a four-link causal chain:

1. **RSSM fidelity gap** (Section 4.1): dt=2 prediction errors are 2.7–3.9× those at dt=1.
2. **Trained dt_embedding enables dt-specific value learning** (Sections 4.3, 4.7): As the embedding differentiates (cossim < 0.4), the critic forms dt-conditioned estimates grounded in inaccurate predictions.
3. **Delusional critic** (Sections 4.4, 4.9): $R_{\text{norm}} > 1.5\times$; value distribution right-tail inflated 3.2×.
4. **Actor degradation and collapse** (Section 4.8): Entropy rises; eval returns decline monotonically.

Each link is independently verified. Removing the dt-embedding (link 2) breaks the chain entirely: R_{norm} stays below 1.0× and no collapse occurs across 9 runs.

5.2 Why This Is Specific to World-Model RL

Model-free temporal abstraction (action repeats, options) does not suffer from this failure mode because there is no world model to exploit. The critic in model-free settings estimates values from *real* environment transitions. In world-model RL, the critic estimates values from *imagined* transitions, and when the imagination is systematically biased at dt>1, the critic's dt-specific estimates become delusional.

5.3 The 1.5× Diagnostic Threshold

The normalized v/e overshoot ratio $R_{\text{norm}} > 1.5\times$ consistently precedes collapse across three environments with different dynamics, reward scales, and baseline performance levels. The baseline v/e calibration curves are similar across environments (asymptotic 65–70%), and the 1.5× threshold separates all 9 collapsed runs from all 9 stable runs with zero misclassification. We propose this metric as a practical, environment-agnostic diagnostic for detecting delusional critic onset in temporally-abstract world-model RL.

5.4 Recommendation: Do Not Jointly Train dt_embedding

Our results provide a clear negative recommendation: **do not jointly train duration embeddings with the value function under current RSSM architectures**. The trained embedding enables the critic to form dt-specific value estimates, but the RSSM's fidelity gap at dt>1 means these estimates are systematically optimistic. Until world model prediction quality at multi-step transitions substantially improves, dt-conditioning is a liability.

Possible mitigation directions include: (1) improving RSSM multi-step prediction fidelity, perhaps through hierarchical latent models; (2) value function regularization that explicitly penalizes dt-conditioned overestimation; (3) conservative value estimation techniques adapted from offline RL; and (4) decoupled training schedules where the dt-embedding is trained on world-model accuracy signals rather than critic gradients.

6 Conclusion

We presented a systematic causal diagnosis of why temporal abstraction fails in world-model reinforcement learning. Through factorial ablation across three DeepMind Control environments with multi-seed replication (9/9 collapse under trained dt-embedding, 9/9 stable under removed embedding), we identified a precise mechanism: trained duration embeddings enable the critic to exploit the RSSM's fidelity gap at dt>1, producing systematic value overestimation that drives actor degradation and performance collapse. The normalized value/eval overshoot ratio ($R_{\text{norm}} > 1.5\times$) serves as a reliable, environment-agnostic early warning diagnostic. Our findings deliver an actionable architectural recommendation: do not jointly train duration embeddings with the value function in current RSSM-based world models.

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